

CHIRAG MODI

WebRTC Engineer | Angular, NodeJS, Python, GoLang | Kubernetes, AWS, WebRTC

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🌐 <https://modi2207.github.io/> </> leetcode.com/u/chiragmodi2207

Education

IIT Gandhinagar

Master of Technology in Computer Science and Engineering

Gandhinagar, Gujarat

CGPA: 9.36/10.0, July 2023 – June 2025

The Maharaja Sayajirao University of Baroda

Bachelor of Engineering in Computer Science and Engineering

Vadodara, Gujarat

CGPA: 3.92/4.0, July 2017 – July 2021

Work Experience

Wrktalk

Software Engineer | NestJS, WebRTC, Redis, Mediasoup, Kubernetes

April 2026 – Present

Mumbai, India

- Architected a multi-pod WebRTC (mediasoup/NestJS) architecture, scaling calls from **20 to 50 participants** using pipe transports, **gRPC** inter-pod signaling, and **Redis** for distributed locking and atomic call-state management.
- Improved call quality and end-to-end latency by implementing **simulcast** encoding with adaptive spatial/temporal layer switching per viewer, and parallelizing signaling flows with async NestJS patterns

Humanyse.ai

Software Engineer - Freelance | LiveKit, NodeJS, GCP, postgresSQL, Prisma

July 2026 – August 2026

Dubai, UAE

- Built and deployed a candidate video-recording microservice for an online assessment platform on **GCP Cloud Run** with secure **RS256 service-to-service auth**.
- Engineered the record-to-playback pipeline using LiveKit Egress to capture assessment sessions in a private GCS bucket, with signed callbacks and short-lived signed playback URLs.

Kapidhwaj AI

Founding Engineer | GCP, FFmpeg, WebRTC, NodeJS, RTSP, ONVIF

May 2024 – Feb 2025

Research Park IIT GN, India

- Built a NodeJS app with ONVIF to manage IP cameras and stream RTSP via FFmpeg with **~800ms latency**.
- Engineered WebRTC communication between user apps and private GCP media servers via TURN in a VPC, achieving **~800ms latency**, **< 2% packet loss**, and stable connectivity across **300+ concurrent sessions**.

WiseDV India

Software Engineer | Angular, NodeJS, WebRTC, AWS, Electron, Mediasoup, Redis, STT

July 2021 – July 2023

San Diego, USA

- Built a scalable WebRTC conferencing app with Mediasoup, handling **500+ users** per session with **~200ms latency**.
- Designed gRPC-based microservices for media and signaling handling over **50K** RPC calls per minute under peak load.
- Configured AWS Auto Scaling Groups for each microservice type, and built a custom NodeJS load balancer to dynamically adjust group capacities based on real-time traffic load using AWS SDK.
- Integrated Google Cloud STT API for live captions and implemented meeting recording in the Electron-based PC app.
- Built a Redis Pub/Sub logging system ingesting **~ 25K logs/sec** to extract insights on session quality and user activity.

Projects

Real-Time Voice-Controlled AI Assistant | LiveKit, Agentic AI, TTS & STT, NodeJS, SIP Trunking

In Progress

- Designing a real-time voice-driven AI assistant using **LiveKit WebRTC rooms** to stream live audio from browser clients and route transcribed speech into a **LangChain/LangGraph-based agent** for conversational task execution.
- Architecting extensibility for **telephony-based voice interaction** by leveraging LiveKit's **SIP trunking and SIP-to-WebRTC bridge**, enabling PSTN callers to participate in AI-assisted workflows as room participants.

eBPF-Based Code Profiling in Kubernetes Deployments | Perforator, Kubernetes, eBPF | Code 🐙

April 2025

- Integrated Yandex's Perforator (eBPF-based code profiler) into the Kubernetes environment to capture function-level execution metrics and call stack samples with **< 3% CPU overhead**.
- Analyzed an open-source DNS resolver to identify hot paths via call stack sampling and optimized compute-bound workloads, reducing **CPU usage by up to 35%**.

Instant Payment Gateway | *HLD & LLD, GoLang, gRPC, ELK, MySQL, NGINX, Kubernetes* | Code April 2024

- Developed a distributed UPI payment gateway system capable of handling up to **10,000 concurrent payment requests**, processing each within an average **latency of 300ms**.
- Implemented the Centralized Coordinator algorithm for mutual exclusion, configured NGINX load balancing, containerized all microservices using Docker, and deployed them on a Kubernetes cluster.

VOIP Call Application | *Angular, NodeJS ELK Stack, WebRTC, Ejabberd XMPP* | Code May 2023

- Designed and implemented a peer-to-peer video call application utilizing WebRTC and XMPP signaling server.
- Integrated Elasticsearch to enable fast username search across **1M+ user records** with response times under **100ms**.

Technical Skills

Languages: C, Python, Java, JavaScript, GoLang

Developer Tools: AWS, GCP, Git and GitHub, Docker, Kubernetes, Wireshark

Technologies/Frameworks: MEAN Stack, NestJS, WebRTC, Electron, Mediasoup, Redis, LiveKit

Databases: MySql, MongoDB, Elastic Search

Course-work: Parallel and distributed systems, Advanced Computer Network, Network and Security, Algorithms

Achievements and ExtraCurricular Activities

- Appointed as Sanchar Mitra by DoT to promote citizen-centric telecom services.
- Earned a **6-star badge** in problem-solving on **HackerRank** and solved more than **850** problems on **LeetCode**
- Secured **AIR 543** in GATE CSE 2023 among **80k candidates**.